

Problem Set 7

Instructor: Robert Rand

1 Probability Theory 1

Problem 1: You walk into Ye Olde Dice Tavern in South Loop to play a game of dice against a random stranger. You know that only one in a thousand dice players use loaded dice, which have a 90% chance of coming up 6 (everyone else uses normal dice). Lo and behold, your opponent's first four rolls all come up heads. What is the probability that she is using loaded dice?

Problem 2: Suppose you roll 2 dice. Are the following events positively correlated, negatively correlated or independent? Show your work.

1. The number on the first die is odd and the total is even.
2. The first die shows a 6 and the absolute difference between the dice is 1.
3. The absolute difference between the dice is 2 and the total is 10.
4. The second die shows a 3 and the difference between the two dice is 6.
5. The first die shows a 4 and the total is 10 or greater.

Problem 3: In a class of 58 people, all of whom were born in the year 2000, what is the probability that exactly 2 were born on Jan 1st, 2000? (Note: The year 2000 was a leap year.)

Problem 4: In a simplified game of Blackjack, we assign values to cards as follows: For each number card i , $V(i) = i$, and for face cards $V(J) = V(Q) = V(K) = 10$, and $V(A) = 11$. What is the expected value of a hand of two cards?